Sustainability in Artificial Intelligence - Towards a Green AI Reference Model

Sebastian Weber[1], Achim Guldner[1], Lejla Begic Fazlic[1], Guido Dartmann[1], Stefan Naumann[1]

Abstract: The interest in Green Artificial Intelligence (AI) is growing as AI research is increasingly focusing on and taking into account environmental sustainability. This paper aims to clarify and emphasize the distinction between terms like sustainable AI, Green AI, Green by AI, and Green in AI, highlighting their importance in the context of environmentally responsible AI practices. We find that existing Green Software reference models are insufficient for meeting the unique requirements of Green AI. Thus, we argue that a tailored Green AI reference model is needed to guide and promote environmentally responsible practices in the field of AI, addressing the special considerations associated with Green AI.

Keywords: Green AI, Sustainable AI, Reference Model, Resource and Energy Consumption

1 Introduction

The use of Artificial Intelligence (AI), and especially its sub-area Machine Learning (ML), increasingly plays a central role in all areas of life and brings many advantages to various application domains [Ma23]. Given the accelerating effects of climate change, it has also become increasingly urgent for every sector to undertake measures to reduce its greenhouse gas emissions, both to mitigate its contribution to this global challenge and to work towards a sustainable future. While it is true that enhancing efficiency and sustainability in domain-specific application areas can be achieved through the optimization with AI and ML-based systems [Ro22; Vi20], equal importance should be placed on constructing these systems in a manner that they adhere to sustainability specifications, as described by Naumann et al. [Ke18; Na11; Na20], without becoming resource drivers.

Schwartz et al. [Sc20] already pointed out that there are two research areas of AI, one that focuses on improving the outcome in the form of accuracy (which they call ’Red AI’) and one that tries to balance the outcome and the impact on the environment (which they call ’Green AI’). There is evidence that the last percentage points in a ’Red AI’ application use a significant amount of energy [Se22], e. g., using twice the energy to gain the last percentage point in accuracy. Current work by Verdecchia et al. [Ve23] showed that the interest in Green AI is rising, but mostly focused on the training phase of ML algorithms (neglecting

[1] Institute for Software Systems, Environmental Campus Birkenfeld, Trier University of Applied Sciences, 55765 Birkenfeld, Germany {seb.weber,a.guldner,l.begic,g.dartmann,s.naumann}@umwelt-campus.de

the rest of the AI pipeline). Despite resulting in promising energy savings of over 50 %, these findings are only rarely transferred to the industry and important stakeholders.

Besides the energy consumption necessary to provide the high compute power, several other challenge arise in the field of Green AI along the life-cycle of an AI-based system. The systems usually require large amounts of data for (re-)training that, in turn, need an infrastructure for collection, transfer, management, and storage, all of which require hardware resources and power to operate (e. g., as explored in Wu et al. [Wu21]). Furthermore, the rapid evolution of the systems often requires the replacement of hardware components, which also requires resources. This aspect was addressed, e. g., by Luccioni et al. [Lu23], who include the embodied emissions associated with the materials and processes involved in the hardware production. However, as pointed out by Bashir et al. [Ba23], integrating embodied emissions into the calculation of the overall footprint can overstate the possible carbon reductions in the hardware production chain and thus lower the incentives for directly optimizing the operational carbon.

To address these issues of AI-based systems, it is necessary to enable the stakeholders of such systems to design, create, operate, and use them in a way that their impact on the environment is as small as possible. To do so, a reference model is required that structures research, helps the industry to understand the issues and make them transparent in their systems, and provides recommendations, processes, and tools to reach the goals of assessing and minimizing the environmental impact of AI-based systems. The objective of this paper is to establish the context and provide a definition for a *Green AI Reference Model*. The section on related work outlines the contemporary understanding of Green and Sustainable AI, Green AI, and Green Software Engineering. It also presents the findings of an exploratory literature review that we undertook with the aim of identifying and categorizing existing models from research, focused on the eco-sustainability of AI-based systems. In the "Approach and Definitions" section, we clarify the concept of "Green AI", explain our understanding of a reference model, and define the term "Green AI Reference Model". Subsequently, we categorize the Green AI reference model. To conclude, we summarize our findings and offer projections on the potential form of this model. The overarching goal of our work is the creation and continuous improvement of the reference model.

2 Related Work

Comparing the research areas of Green and Sustainable Software and Green and Sustainable AI with each other, the following Fig. 1 juxtaposes the amounts of research papers found in the context of the two fields from the first contributions in 2009. While there are papers containing the term "Sustainable Software" before 2009, like [HL08], these sources usually use the term sustainable in the sense of "lasting" or "durable" and refer to issues like modular, interoperable, and reusable software. Of course, this is also one aspect of sustainability, but before 2009, no papers were found that considered ecological sustainability. Therefore, 2009 was set as the start of the interval. For 2023, the number was extrapolated by multiplying the results already available at the end of April by three (dashed lines).

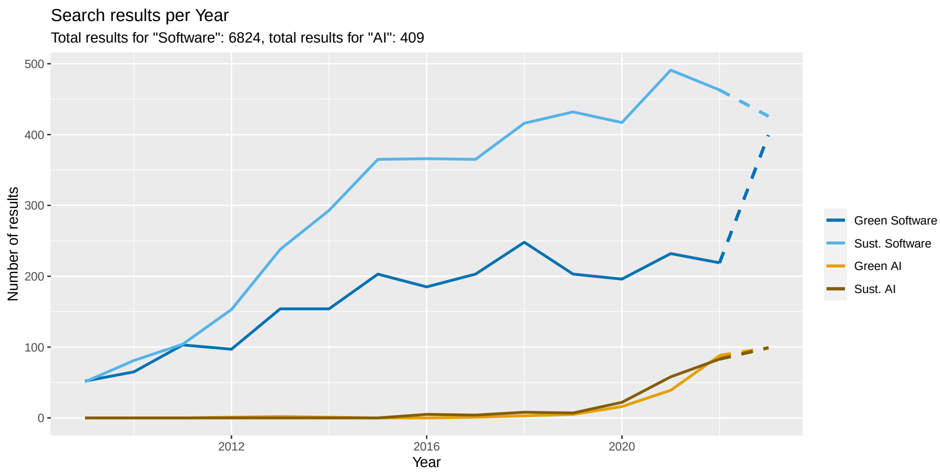

Fig. 1: Google scholar results for green and sustainable software and green and sustainable AI

2.1 Green and Sustainable AI

Sustainability is a complex concept that can be organized into different aspects, commonly referred to as dimensions or pillars, depending on the definition. These dimensions, including ecological, social, and economic, help categorize and understand the various components of sustainability, which was already summarized for the field of software, e. g., by Calero et al. [Ca21] and by van Wynsberghe [Va21] for the field of AI. From this, we derive the concept of Sustainable AI to entail achieving balance across all dimensions of sustainability, whereas Green AI – as a part of Sustainable AI – focuses on ecological sustainability.

There are several different approaches to make AI “greener”, i. e., to optimize AI-based software systems, such that their environmental impact is as low as possible. As Verdecchia et al. [Ve23] point out, there are different approaches towards Green AI, for example measuring the energy consumption during training or inference. This not only makes the energy requirements transparent for developers, users, or customers, but also aides developers if they want to optimize their systems. The AI community *Huggingface* added a CO_2 emission field to its models[2]. Although attempts have been made to demonstrate and compare the energy usage and CO_2 footprint of AI models (e. g., [Ge22b; Gu21; He20]), it remains challenging to make meaningful comparisons in actual practice. For instance, the process of incorporating the CO_2 equivalent emitted by a model during its training into a meaningful comparison with another system requires a comprehensive understanding of the specific hardware used, the energy mix during that time, and other pertinent factors.

[2] `https://huggingface.co/docs/hub/model-cards-co2` [2023-05-01]

Looking deeper in the current discussion around large language model (LLM), e. g., authors in [Ge22a; RA22] analyzed the performance and carbon footprint for BERT-based approaches, light models, and hybrid (symbolic and ML approaches) when they are used in very specific and vertical domains. In a series of experiments to examine how LLM lose value, they document clear disadvantages in terms of the necessary computational sources and energy consumption. The classification results in particular domains show that using LLMs does not demonstrate a substantial difference compared to other approaches even when they are used in a specific language domain. Implementation of a symbolic approach that combines domain specific linguistic knowledge into the feature representation enables achieving good performance. The carbon footprint and energy consumption are measured for different models during testing, training, and validation. The results by above mentioned authors show that the "light"models and symbolic AI actually achieved energy efficiency savings of around 75 % when compared to BERT-based models. This corresponds to a comparable reduction in both cost and carbon footprint. It is possible to have NLP systems that are fast, have a high performance, and a smaller ecological footprint. These results show that it is necessary to systematically examine optimization possibilities, which can be assisted with a structured model. They also show that measured values (like 75 % savings in this case) are valuable but can also be deceiving when they cannot be interpreted well, without comparisons, metrics, etc.

2.2 Green AI and Green Software Engineering

AI-based systems are, of course, also software systems. Hence, Green AI is a sub-field that belongs to the broader category of Green Software Engineering (GSE). These two fields exhibit certain similarities, in particular, when it comes to the need for a platform that delivers the AI-based service (e. g., a website for the user interface). GSE also faces challenges in standardizing its terminology, which has resulted in the interchangeable use of terms like Green Coding and Sustainable Software Engineering (SSE). This was, e. g., addressed by Calero and Piattini [CP17]. Nonetheless, due to the unique characteristics of AI, such as the ability to “learn” rather than being strictly algorithm-based and its black-box nature, it necessitates a distinctive approach.

2.3 Green AI Reference Models in the Literature

In our research, we encountered difficulties finding a (reference) model that provides a comprehensive overview of the Green AI topic. Therefore, we scanned the literature, to assess if (1) any approaches for a reference model in the area of Green AI already exist and (2) which reference models are available for GSE that could be applied or transferred and adapted to the area of Green AI. We began by formulating search queries that took into account the various terms commonly used in the field and searched the Google Scholar[3] and dblp[4] databases. To ensure a comprehensive search, we included acronyms associated with AI and GSE in our queries. This helped us capture a broader range of relevant publications. One of our key objectives was to identify papers that discussed or proposed a reference model in the context of our research topic. We specifically looked for works that provided a structured framework or model for understanding and implementing concepts related to AI and GSE. Since the comprehensive results and search queries would exceed the scope of this publication, we provide the lists as supplementary material at `https://gitlab.rlp.net/green-software-engineering/towards-a-green-ai-reference-model-supplementary-material`.

The scan shows that several, differently named models (like process, life-cycle, or measurement models), frameworks, etc. exist. However, especially one model (i. e., [Na11]) can be viewed as a state-of-the-art reference model in the area of Green Software. Despite its continued development [Ke13; Ke15; Ke18; Na15], it is only partially applicable for Green AI due to several reasons: (1) It lacks an extension to address the unique requirements of AI applications due to their learning nature and (usually) black-box characteristic. (2) It does not incorporate current research in the sub-field of Green AI.

3 Approach and Definitions

Based on the exploratory literature scan and previously known papers, we compiled an overview of the state of research in the area of Green AI, and especially models, recommendations, and tools that aid stakeholders to analyze, evaluate, and optimize the ecological sustainability of their AI-based systems. The overarching goal is to create a reference model for Green AI, which we delineate in the following. To approach this goal, we have compiled definitions for the most important terms, a model hierarchy, and initial considerations about the possible contents of the model.

3.1 Green By AI vs. Green AI vs. Green in AI

Ongoing discussions revolve around the terms “Sustainable AI”, “Ecologically Sustainable AI”, and “Green AI”. In addition to this, there is another conversation surrounding the term “Green (by/in) AI”. In literature and research, the term “Green AI” encompasses various interpretations depending on the context it is used in: One interpretation of “Green AI” involves the endeavor to make AI technology itself more environmentally friendly and energy-efficient (e. g., as discussed by [Sc20]). This perspective focuses on reducing the carbon footprint and resource consumption associated with AI systems throughout their life cycle, encompassing stages such as design, training, deployment, and operation.

[3] https://scholar.google.com/
[4] https://dblp.uni-trier.de/db/

Alternatively, “Green AI” is also used to describe applications that leverage AI[5] to promote environmentally friendly practices and make processes more ecologically sustainable. This aspect is sometimes also referred to as “Green by AI” (e. g., in [Wa21]). These applications utilize AI algorithms and techniques to optimize resource allocation, minimize waste, or enhance efficiency across diverse domains such as energy production, transportation, agriculture, and more (Rolnick et al. [Ro22] produced an extensive list, with ways to apply AI/ML to tackle climate change).

In this paper, we aim to establish a shared understanding and facilitate effective communication regarding the various dimensions of “Green AI”. To promote a comprehensive understanding of the term, similar to van Wynsberghe’s [Va21] approach in defining sustainable AI, we propose the following definition, aligning it with the definition of “Green Software” by Naumann et al. [Na11].

Definition 1 (Green AI) *Green AI refers to AI-based systems, whose direct and indirect negative impacts on the environment that result from development (e. g. training), deployment (e. g. MLOps), usage (e. g. inference) and end-of-life of the systems are minimal* “Green in AI” *and/or which have a positive effect on ecologically sustainable development* “Green by AI”.

Acknowledging that the term “AI for Green” is used interchangeably with “Green by AI” as employed, e. g., in [Wy22]. It should be pointed out that “Green in AI” and “Green by AI” approaches define a spectrum in which the AI-based systems can be ranked. This means that a “Green in AI” system can be applied onto a “Green by AI” problem. However, the AI use case should be considered regarding its ecological impact as well. Similarly, this principle applies to “Green by AI”. While it is not obligatory for “Green by AI” applications to adhere to environmentally friendly AI best practices, it is crucial to recognize that the overall conservation of resources should not be compromised by utilizing AI. Consequently, it is always necessary to consider whether a non-AI solution exists that can effectively address the problem while requiring fewer resources.

3.2 Reference Model

In software engineering, reference models are important tools to structure and guide research as well as help practitioners to quickly understand an area. According to Fettke and vom Brocke [FB19] however, there is still no generally accepted definition for reference models in computer science. We therefore first define what we understand as a reference model in the context of this work and give some general examples, before we define the term *Green AI Reference Model*.

[5] e. g., in `https://www.green-ai-hub.de/` [2023-04-30]

3.2.1 Definition of 'Reference Model'

For this paper, we build upon a modified definition from MacKenzie et al. [Ma06].

Definition 2 (Reference model) *A Reference Model serves as an abstract framework for comprehending the primary entities and their significant interrelationships within a particular environment or domain. It facilitates the development of specific reference architectures or concrete architectures by adhering to consistent standards or specifications relevant to that environment. More specifically, it should comply with the following requirements: A reference model*

- *helps stakeholders (researchers, developers, users, etc.) to understand the big picture of a topic by structuring its main components, e. g., by classifying common terminologies,*
- *provides guidance to create new, or categorize more specific, models or to compare frameworks, and*
- *helps to identify potential and gaps in research.*

But it is independent of specific standards, technologies, implementations, or other concrete details.

Since at present, the field of Sustainable AI is rapidly evolving, in addition to these requirements, the reference model should be able to adapt to these changes and be updated to include emerging research results.

3.2.2 Green AI Reference model

Accordingly, we define the "Green Artificial Intelligence Reference Model" as follows.

Definition 3 (Green AI Reference Model) *The Green AI Reference Model is an abstract framework for understanding the principal entities and their significant interrelationships that impact the ecological sustainability of AI-based software systems. It enables the development of specific models, methods, procedures, or architectures, employing unified terms, definitions, standards, and specifications. These collectively ensure that the life-cycle processes of AI-based systems, from creation over usage to disposal, are carried out in a manner that aligns with the principles of Green AI (see Def. 1).*

Furthermore, the reference model should address the following aspects:
Supporting stakeholders*: It should classify and sort common definitions and the partially ambiguous terminology (such as Green AI, Green By AI, Sustainable AI, etc.), e. g., in form of a glossary or taxonomy. Furthermore, the stakeholders' perspectives should be included*

into each model component.
Modeling*: It should provide guidance to create new or categorize existing, yet more specific models and allow the comparison of frameworks. In this regard, it should allow state-of-the-art frameworks to be adapted to support Green AI.*
Enable research*: It should allow scientists to identify gaps in research, e. g., by outlining the critical elements (like measurements, transparency of the resource consumption, etc.), so that areas that require more investigation become obvious. Hence, it should allow the creation of more effective and efficient Green AI models.*
Ecological sustainability*: In addition to the general goals of the reference model, it must enable the analysis and evaluation of the AI-based systems in terms of issues that cause resource and energy consumption, which triggers impact the consumption, and how the systems can be optimized to reduce these impacts.*

4 Reference Modeling and Discussion

Fig. 2 shows a conceptual categorization of exemplary and existing state-of-the-art models that can be derived from or integrated into the Green AI Reference Model. This illustrates where the reference model should be placed in the model hierarchy of Green Software.

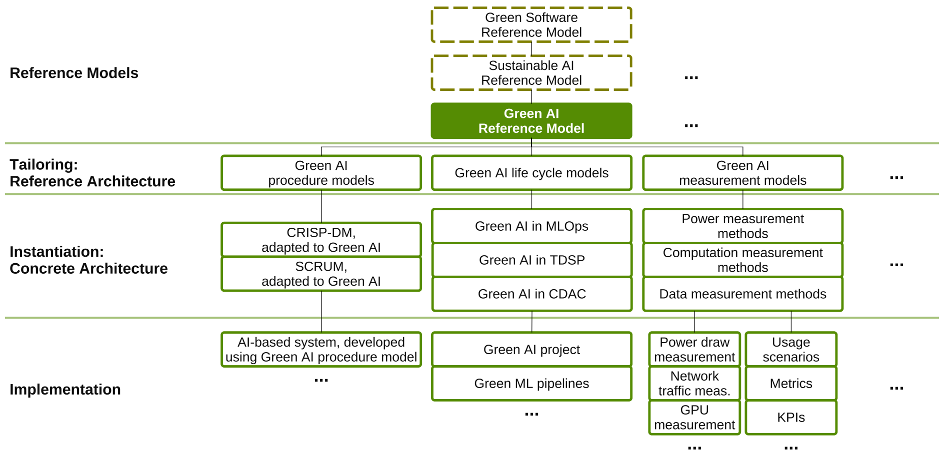

Fig. 2: Reference model hierarchy, instantiation/tailoring, and examples

The model itself can be seen as a tailored Sustainable AI Reference Model (which should also contain models for the other pillars of sustainability), which, in turn, can be seen as a tailored Green Software Reference Model, as described by [Na11]. The Green AI Reference Model can then be tailored to include individual reference architectures, like procedure models, life cycle models, measurement models, etc. that each contain more and more concrete models, with the leafs of the hierarchy representing the specific implementations, measurements, ML models, etc.

To elaborate on the model in the future, we propose a two-pronged approach: Through the vertical expansion of further reference architectures, the addition of stakeholders' perspectives, the addition of measurement models, criteria, and indicators, etc. we aim at broadening the hierarchy to include new and existing architectures (top-down). By integrating existing Green Software and Green AI approaches into state-of-the-art models like CRISP-DM [WH00], CRISP-ML(Q) [St21], MLOps [Kr23], TDSP [Mi17], or CDAC [DA22], or AI projects in development (bottom-up), we want to tailor and verify the reference model, detect influencing factors, research gaps, or missing reference architectures. To assess our Green AI Reference Model, we plan to apply it to existing AI-based applications as well as completely new projects and to create surveys for industry and researchers.

5 Conclusion and outlook

The concept of sustainable AI encompasses several dimensions that need consideration during the development of AI-based applications. We highlighted the significance of a reference model for enhancing our comprehension of Sustainable AI and guiding research. Consequently, we conducted an investigation to identify existing models that adhere to the specified criteria for a Green AI Reference Model. However, our search yielded no suitable options, and also the extension to Green Software did not yield a satisfactory model. The absence of a Green AI Reference Model underscores the pressing need for its development.

In the future, we will focus on the creation of the Green AI Reference Model that serves as a framework for guiding the design and implementation of environmentally conscious AI applications. By incorporating the specific requirements of AI-based systems, including their learning nature and the evolving field of Green Artificial Intelligence, a comprehensive Green AI Reference Model could effectively address the sustainability challenges inherent in AI development. The development of such a reference model should provide researchers, developers, and stakeholders with a standardized approach to ensure the integration of sustainable practices into AI systems, thereby promoting environmentally friendly and socially responsible AI innovation. The reference model should be modular, generic, and expandable to include tailored sub-models and ideas like criteria, metrics, measurement methods, etc. We plan the development in a bottom-up and top-down approach, so that it supports and builds upon already established models, like MLOps, CRISP-DM, etc.

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